ResoAI DuckDB Medallion Lakehouse

LIVE EMBEDDED LAKEHOUSE

Complete Database Schema, Star Schema Dimensional Architecture & Live Sample Data Catalog

ENGINE	TOTAL TABLES / VIEWS	TOTAL MANAGED RECORDS	STORAGE ARCHITECTURE	EXECUTION SPEED	CATALOG GENERATED
Embedded DuckDB v2.0	13 Tables	1,619 Records	Medallion (Bronze / Silver / Gold Parquet)	< 200 ms (Sub-second SQL)	October 30, 2026 • 03:38 PM IST

Medallion Architecture Master Inventory

LAYER	CATEGORY	TABLE / VIEW NAME	TOTAL ROWS	COLUMNS	STORAGE FORMAT & PRIMARY ARCHITECTURAL ROLE
GOLD	Fact Table	`fact_employee_allocations`	500	9	Central quantitative fact table recording salary, experience, performance score, leav...
GOLD	Dimension Table	`dim_employee`	500	12	Conformed dimension containing normalized employee biographical, contact, role, skill...
GOLD	Dimension Table	`dim_department`	8	4	Organizational department dimension defining business units, functional departments, ...
GOLD	Dimension Table	`dim_project`	6	6	Project dimension cataloging enterprise client engagements, internal innovation initi...
GOLD	Dimension Table	`dim_holiday`	24	7	Official company holiday dimension calendar defining fixed corporate holidays, floate...
GOLD	Dimension Table (View/Alias)	`company_holidays`	24	7	Sub-millisecond analytical view directly exposing the 2026 FiftyFive Technologies cor...
GOLD	Analytical Aggregation Mart	`gold_dept_summary`	8	8	Pre-computed analytical summary calculating total department headcounts, average/min/...
GOLD	Analytical Aggregation Mart	`gold_city_headcount`	5	4	Geographical distribution rollup showing active workforce headcount, department diver...
GOLD	Analytical Aggregation Mart	`gold_project_staffing`	6	5	Project-level allocation summary showing assigned engineering team size, solution dom...
SILVER	Cleaned & Enriched Entity	`silver_employees`	500	20	Deduplicated, standardized employee master table with cleaned types, resolved departm...
SILVER	Cleaned Entity	`silver_departments`	8	4	Cleaned and verified department master table with canonical names and headquarters/hu...
SILVER	Cleaned Entity	`silver_projects`	6	6	Cleaned project portfolio table containing client classifications, tech stacks, and d...
SILVER	Cleaned Entity	`silver_holidays`	24	7	Cleaned corporate holiday calendar with validated date formats, day names, and leave ...

BRONZE LAYER **Raw Immutable Ingestion Payloads (JSON Archive)** SOURCE: SYNTHETIC_GENERATOR_API

Streaming raw JSON documents into `data/lakehouse/bronze/bronze_hr_raw_latest.json`. Contains raw ingested arrays: **500 employees**, **8 departments**, **6 projects**, and **24 official company holidays**. Pipeline Ingestion Timestamp: `2026-10-30T12:53:23.398497`.

Gold Layer — Star Schema (Central Fact Table)

Quantitative transactions & metric aggregations

1 TABLE

GOLD LAYER **fact_employee_allocations** • Fact Table

500 Rows | 9 COLUMNS

Central quantitative fact table recording salary, experience, performance score, leave balance, and project allocation percentage, joined against dimensional surrogate keys.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **fact_id** `BIGINT` (Primary Key)
- **department_key** `BIGINT` (Foreign Key -> dim_department)
- **salary** `DOUBLE` (Metric (Annual CTC in INR))
- **performance_rating** `DOUBLE` (Metric (1.0 - 5.0 Rating))
- **allocation_percentage** `INTEGER` (Metric (Staffing %))
- **employee_key** `BIGINT` (Foreign Key -> dim_employee)
- **project_key** `BIGINT` (Foreign Key -> dim_project)
- **years_of_experience** `DOUBLE` (Metric (Years))
- **leave_balance** `INTEGER` (Metric (Available Days))

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

FACT_ID	EMPLOYEE_KEY	DEPARTMENT_KEY	PROJECT_KEY	SALARY	YEARS_OF_EXPERIENCE	PERFORMANCE_RATING	LEAVE_BALANCE	ALLOCATION_PERCENTAGE
1	1	2	4	₹649,256.00	1.90	3.60	20	100
2	2	1	3	₹2,474,987.00	12.00	4.70	14	100
3	3	2	1	₹667,052.00	3.20	3.80	24	100
4	4	6	3	₹1,526,006.00	7.90	3.90	15	100

Gold Layer — Star Schema (Core Employee Dimension)

Normalized workforce attributes & skill taxonomy

1 TABLE

GOLD LAYER **dim_employee** • Dimension Table

500 Rows | 12 COLUMNS

Conformed dimension containing normalized employee biographical, contact, role, skill taxonomy, and employment lifecycle attributes.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **employee_key** `BIGINT` (Primary Key (Surrogate))
- **employee_id** `VARCHAR` (Business Key / Natural Key)
- **full_name** `VARCHAR` (Full Name)
- **first_name** `VARCHAR` (First Name)
- **last_name** `VARCHAR` (Last Name)
- **email** `VARCHAR` (Corporate Email Address)
- **phone** `VARCHAR` (Contact Phone Number)
- **designation** `VARCHAR` (Job Title / Corporate Rank)
- **location** `VARCHAR` (Office Hub / Remote Location)
- **skills** `VARCHAR` (Comma-delimited Core Tech Skills)
- **joining_date** `DATE` (Date of Joining)
- **status** `VARCHAR` (Employment Status (Active/Inactive))

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

EMPLOYEE_KEY	EMPLOYEE_ID	FULL_NAME	EMAIL	PHONE	DESIGNATION	LOCATION	SKILLS	JOINING_DATE	STATUS
1	EMP0001	Isaac Bakshi	isaac.bakshi@fiftyfivetech.io	4332181960	Associate Software Engineer	Udaipur	Microservices, FastAPI, Java	2022-09-30	Active
2	EMP0002	Ayushman Chander	ayushman.chander@fiftyfivetech.io	+913890838637	Head of AI	Gurgaon	NLP, Python, RAG, PyTorch, Hugg...	2026-02-10	Active
3	EMP0003	Arunima Ahuja	arunima.ahuja@fiftyfivetech.io	+916542351161	Associate Software Engineer	Bengaluru	Java, Python, Go, Node.js, Micr...	2021-01-31	Active
4	EMP0004	Jagrati Padmanabhan	jagrati.padmanabhan@fiftyfivete...	4078161849	Senior QA Lead	Gurgaon	Automation Testing, Selenium, P...	2020-10-11	Active

Gold Layer — Star Schema (Department & Project Dimensions)

Organizational departments, regional hubs, client accounts & tech stacks

2 TABLES

GOLD LAYER dim_department • Dimension Table

8 Rows | 4 COLUMNS

Organizational department dimension defining business units, functional departments, and primary physical operating locations.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **department_key** `BIGINT` (Primary Key (Surrogate))
- **department_id** `VARCHAR` (Business Key)
- **department_name** `VARCHAR` (Department / Functional Unit Name)
- **location_hub** `VARCHAR` (Primary Operating City Hub)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

DEPARTMENT_KEY	DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_HUB
1	DEP01	Artificial Intelligence	Jaipur
2	DEP02	Engineering	Jaipur
3	DEP03	Product & Design	Gurgaon
4	DEP04	Human Resource	Jaipur

GOLD LAYER dim_project • Dimension Table

6 Rows | 6 COLUMNS

Project dimension cataloging enterprise client engagements, internal innovation initiatives, domains, and assigned technology stacks.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **project_key** `BIGINT` (Primary Key (Surrogate))
- **project_id** `VARCHAR` (Business Key)
- **project_name** `VARCHAR` (Project Name)
- **domain** `VARCHAR` (Industry / Solution Domain)
- **tech_stack** `VARCHAR` (Core Technologies & Frameworks)
- **client_name** `VARCHAR` (Client Account Name)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

PROJECT_KEY	PROJECT_ID	PROJECT_NAME	DOMAIN	TECH_STACK	CLIENT_NAME
1	PRJ01	ResoAI	Enterprise AI / NLP	Python, Ollama, ChromaDB, FastA...	FiftyFive Technologies Internal
2	PRJ02	VisionAI OCR	Computer Vision	Python, PyTorch, OpenCV, Fastif...	HealthCorp Global
3	PRJ03	CloudMigrate	Cloud Infrastructure	AWS, Terraform, Kubernetes, Go	FinTech Alliance
4	PRJ04	RetailPulse	E-Commerce Analytics	React, Node.js, PostgreSQL, Redis	OmniRetail Inc

Gold Layer — Star Schema (Corporate Holiday Calendar)

Official FiftyFive Technologies 2026 holiday calendar dimension and fast-path view

2 TABLES

GOLD LAYER **dim_holiday** • Dimension Table

24 Rows | 7 COLUMNS

Official company holiday dimension calendar defining fixed corporate holidays, floater leaves, weekdays, and holiday classifications.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **holiday_key** `BIGINT` (Primary Key (Surrogate))
- **holiday_id** `VARCHAR` (Unique Holiday Identifier)
- **holiday_name** `VARCHAR` (Official Holiday Name)
- **date** `DATE` (Calendar Date)
- **day** `VARCHAR` (Day of the Week)
- **month** `VARCHAR` (Calendar Month)
- **holiday_type** `VARCHAR` (Classification (Fixed / Floater / Weekend))

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

HOLIDAY_KEY	HOLIDAY_ID	HOLIDAY_NAME	DATE	DAY	MONTH	HOLIDAY_TYPE
1	HOL_NEW_YEAR__2026_2026	New Year - 2026	2026-01-01	Thursday	January	Fixed
2	HOL_MAKAR_SANKRANTI__PONGAL_2026	Makar Sankranti / Pongal	2026-01-14	Wednesday	January	Floater Leave
3	HOL_REPUBLIC_DAY_2026	Republic Day	2026-01-26	Monday	January	Fixed
4	HOL_HOLI_2026	Holi	2026-03-04	Wednesday	March	Fixed

GOLD LAYER **company_holidays** • Dimension Table (View/Alias)

24 Rows | 7 COLUMNS

Sub-millisecond analytical view directly exposing the 2026 FiftyFive Technologies corporate holiday calendar for rapid SQL lookups.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **holiday_key** `BIGINT` (Surrogate Identifier)
- **holiday_id** `VARCHAR` (Unique Holiday Identifier)
- **holiday_name** `VARCHAR` (Holiday Title)
- **date** `DATE` (Holiday Date)
- **day** `VARCHAR` (Day of the Week)
- **month** `VARCHAR` (Month Name)
- **holiday_type** `VARCHAR` (Holiday Classification)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

HOLIDAY_KEY	HOLIDAY_ID	HOLIDAY_NAME	DATE	DAY	MONTH	HOLIDAY_TYPE
1	HOL_NEW_YEAR__2026_2026	New Year - 2026	2026-01-01	Thursday	January	Fixed
2	HOL_MAKAR_SANKRANTI__PONGAL_2026	Makar Sankranti / Pongal	2026-01-14	Wednesday	January	Floater Leave
3	HOL_REPUBLIC_DAY_2026	Republic Day	2026-01-26	Monday	January	Fixed
4	HOL_HOLI_2026	Holi	2026-03-04	Wednesday	March	Fixed

Gold Layer — Analytical Aggregation Marts (Part 1)

Sub-millisecond materialized views for departmental summaries & city distributions

2 TABLES

GOLD LAYER gold_dept_summary • Analytical Aggregation Mart

8 Rows | 8 COLUMNS

Pre-computed analytical summary calculating total department headcounts, average/min/max salary distributions, experience, and performance.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **department_name** VARCHAR (Department Name)
- **location_hub** VARCHAR (Location Hub)
- **total_headcount** BIGINT (Aggregated Employee Count)
- **avg_salary** DOUBLE (Average Annual Salary (INR))
- **min_salary** DOUBLE (Minimum Annual Salary (INR))
- **max_salary** DOUBLE (Maximum Annual Salary (INR))
- **avg_performance_rating** DOUBLE (Average Rating (1-5))
- **avg_years_experience** DOUBLE (Average Experience (Years))

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

DEPARTMENT_NAME	LOCATION_HUB	TOTAL_HEADCOUNT	AVG_SALARY	MIN_SALARY	MAX_SALARY	AVG_PERFORMANCE_RATING	AVG_YEARS_EXPERIENCE
Product & Design	Gurgaon	71	₹1,363,616.75	₹621,493.00	₹2,086,923.00	4.03	6.30
Data Analytics	Udaipur	67	₹1,464,269.67	₹628,851.00	₹2,087,739.00	3.91	6.80
Quality Assurance	Jaipur	64	₹1,123,291.48	₹631,593.00	₹1,644,856.00	3.90	5.00
Engineering	Jaipur	64	₹1,706,846.27	₹641,473.00	₹2,533,701.00	3.92	8.20

GOLD LAYER gold_city_headcount • Analytical Aggregation Mart

5 Rows | 4 COLUMNS

Geographical distribution rollup showing active workforce headcount, department diversity, and compensation across company locations.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **location** VARCHAR (City Hub / Location)
- **employee_count** BIGINT (Total Headcount in City)
- **departments_present** BIGINT (Distinct Departments Operating)
- **avg_salary** DOUBLE (Average Salary in City (INR))

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

LOCATION	EMPLOYEE_COUNT	DEPARTMENTS_PRESENT	AVG_SALARY
Gurgaon	104	8	₹1,399,766.89
Remote	103	8	₹1,418,707.34
Bengaluru	99	8	₹1,518,278.63
Jaipur	99	8	₹1,447,073.85

Gold Layer — Analytical Aggregation Marts (Part 2)

Project allocation summary & assigned team performance metrics

1 TABLE

GOLD LAYER **gold_project_staffing** • Analytical Aggregation Mart

6 Rows | 5 COLUMNS

Project-level allocation summary showing assigned engineering team size, solution domain, and average team performance metrics.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **project_name** `VARCHAR` (Project Name)
- **client_name** `VARCHAR` (Client Name)
- **avg_team_rating** `DOUBLE` (Average Team Rating (1-5))
- **domain** `VARCHAR` (Domain Name)
- **assigned_team_size** `BIGINT` (Number of Assigned Employees)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

PROJECT_NAME	DOMAIN	CLIENT_NAME	ASSIGNED_TEAM_SIZE	AVG_TEAM_RATING
VisionAI OCR	Computer Vision	HealthCorp Global	92	3.93
ResoAI	Enterprise AI / NLP	FiftyFive Technologies Internal	88	4.00
CloudMigrate	Cloud Infrastructure	FinTech Alliance	87	3.94
RetailPulse	E-Commerce Analytics	OmniRetail Inc	80	3.94

Silver Layer — Cleaned & Enriched Entity (Employees)

Deduplicated, schema-validated Parquet dataset with 20 attributes & entity inspection

1 TABLE

SILVER LAYER **silver_employees** • Cleaned & Enriched Entity

500 Rows | 20 COLUMNS

Deduplicated, standardized employee master table with cleaned types, resolved department mappings, and Parquet storage backing.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **employee_id** VARCHAR (Natural Key)
- **last_name** VARCHAR (Last Name)
- **email** VARCHAR (Standardized Email)
- **department_id** VARCHAR (Department ID)
- **designation** VARCHAR (Standardized Title)
- **joining_date** DATE (Casted Date)
- **years_of_experience** DOUBLE (Experience (Float))
- **status** VARCHAR (Lifecycle Status)
- **project_name** VARCHAR (Assigned Project Name)
- **leave_balance** INTEGER (Balance (Integer))
- **first_name** VARCHAR (First Name)
- **full_name** VARCHAR (Combined Full Name)
- **phone** VARCHAR (Phone Number)
- **department_name** VARCHAR (Department Name)
- **location** VARCHAR (Standardized Location)
- **salary** DOUBLE (Annual CTC (Float))
- **skills** VARCHAR (Skills String)
- **project_id** VARCHAR (Assigned Project ID)
- **performance_rating** DOUBLE (Rating (Float))
- **processed_at** TIMESTAMP WITH TIME ZONE (Silver Ingestion Timestamp)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

EMPLOYEE_ID	FULL_NAME	DEPARTMENT_NAME	DESIGNATION	LOCATION	SALARY	YEARS_OF_EXPERIENCE	SKILLS	STATUS	PROJECT_NAME
EMP0001	Isaac Bakshi	Engineering	Associate Software Engineer	Udaipur	₹649,256.00	1.90	Microservices, FastAPI, Java	Active	RetailPulse
EMP0002	Ayushman Chander	Artificial Intelligence	Head of AI	Gurgaon	₹2,474,987.00	12.00	NLP, Python, RAG, PyTorch, Hugg...	Active	CloudMigrate
EMP0003	Arunima Ahuja	Engineering	Associate Software Engineer	Bengaluru	₹667,052.00	3.20	Java, Python, Go, Node.js, Micr...	Active	ResoAI
EMP0004	Jagrati Padmanabhan	Quality Assurance	Senior QA Lead	Gurgaon	₹1,526,006.00	7.90	Automation Testing, Selenium, P...	Active	CloudMigrate

RECORD ENTITY INSPECTOR (ROW 1 • ALL 20 ATTRIBUTES):

EMPLOYEE_ID **EMP0001**	FIRST_NAME **Isaac**	LAST_NAME **Bakshi**	FULL_NAME **Isaac Bakshi**
EMAIL **isaac.bakshi@fiftyfivetech.io**	PHONE **4332181960**	DEPARTMENT_ID **DEP02**	DEPARTMENT_NAME **Engineering**
DESIGNATION **Associate Software Engineer**	LOCATION **Udaipur**	JOINING_DATE **2022-09-30**	SALARY **₹649,256.00**
YEARS_OF_EXPERIENCE **1.90**	SKILLS **Microservices, FastAPI, Java**	STATUS **Active**	PROJECT_ID **PRJ04**
PROJECT_NAME **RetailPulse**	PERFORMANCE_RATING **3.60**	LEAVE_BALANCE **20**	PROCESSED_AT **2026-10-30 12:53:23.476991+05:30**

Silver Layer — Cleaned Reference Parquet Tables

Standardized operational tables for departments, projects, and holidays

3 TABLES

SILVER LAYER **silver_departments** • Cleaned Entity

8 Rows | 4 COLUMNS

Cleaned and verified department master table with canonical names and headquarters/hub assignments.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **department_id** VARCHAR (Department Key)
- **department_name** VARCHAR (Department Name)
- **location_hub** VARCHAR (Operating Hub)
- **processed_at** TIMESTAMP WITH TIME ZONE (Silver Pipeline Timestamp)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_HUB	PROCESSED_AT
DEP01	Artificial Intelligence	Jaipur	2026-10-30 12:53:23.445626+05:30
DEP02	Engineering	Jaipur	2026-10-30 12:53:23.445626+05:30
DEP03	Product & Design	Gurgaon	2026-10-30 12:53:23.445626+05:30
DEP04	Human Resource	Jaipur	2026-10-30 12:53:23.445626+05:30

SILVER LAYER **silver_projects** • Cleaned Entity

6 Rows | 6 COLUMNS

Cleaned project portfolio table containing client classifications, tech stacks, and domain categorization.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **project_id** VARCHAR (Project Key)
- **project_name** VARCHAR (Project Name)
- **domain** VARCHAR (Domain)
- **tech_stack** VARCHAR (Tech Stack)
- **client_name** VARCHAR (Client Name)
- **processed_at** TIMESTAMP WITH TIME ZONE (Silver Pipeline Timestamp)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

PROJECT_ID	PROJECT_NAME	DOMAIN	TECH_STACK	CLIENT_NAME	PROCESSED_AT
PRJ01	ResoAI	Enterprise AI / NLP	Python, Ollama, ChromaDB, FastA...	FiftyFive Technologies Internal	2026-10-30 12:53:23.462442+05:30
PRJ02	VisionAI OCR	Computer Vision	Python, PyTorch, OpenCV, Fastif...	HealthCorp Global	2026-10-30 12:53:23.462442+05:30
PRJ03	CloudMigrate	Cloud Infrastructure	AWS, Terraform, Kubernetes, Go	FinTech Alliance	2026-10-30 12:53:23.462442+05:30
PRJ04	RetailPulse	E-Commerce Analytics	React, Node.js, PostgreSQL, Redis	OmniRetail Inc	2026-10-30 12:53:23.462442+05:30

SILVER LAYER **silver_holidays** • Cleaned Entity

24 Rows | 7 COLUMNS

Cleaned corporate holiday calendar with validated date formats, day names, and leave categorization.

COLUMN SPECIFICATIONS & ARCHITECTURAL ROLES:

- **holiday_id** VARCHAR (Holiday Key)
- **holiday_name** VARCHAR (Holiday Name)
- **date** DATE (Parsed Date)
- **day** VARCHAR (Day of Week)
- **month** VARCHAR (Month)
- **holiday_type** VARCHAR (Classification)
- **processed_at** TIMESTAMP WITH TIME ZONE (Silver Pipeline Timestamp)

LIVE SAMPLE DATA PREVIEW (FIRST 4 ROWS):

HOLIDAY_ID	HOLIDAY_NAME	DATE	DAY	MONTH	HOLIDAY_TYPE	PROCESSED_AT
HOL_NEW_YEAR__2026_2026	New Year - 2026	2026-01-01	Thursday	January	Fixed	2026-10-30 12:53:23.502069+05:30
HOL_REPUBLIC_DAY_2026	Republic Day	2026-01-26	Monday	January	Fixed	2026-10-30 12:53:23.502069+05:30
HOL_HOLI_2026	Holi	2026-03-04	Wednesday	March	Fixed	2026-10-30 12:53:23.502069+05:30
HOL_INDEPENDENCE_DAY_2026	Independence Day	2026-08-15	Saturday	August	Weekend	2026-10-30 12:53:23.502069+05:30